

Number Systems, Representation, and Arithmetic Hardware

Week 2 — What the Bits Mean, and How Hardware Computes With Them

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Where We Left Off (Week 1)

Week 1 was about *speed*

- ▶ Architecture (the interface) vs. organization (the implementation).
- ▶ The performance equation: $\text{CPU time} = \text{IC} \times \text{CPI}/f$ — three independent levers.
- ▶ Amdahl's Law: the unimproved fraction sets a hard ceiling.

The pivot today

The equation assumed we can *count* instructions and *measure* CPI. But the CPU cannot execute anything until the **data** it works on is represented in bits — and arithmetic is implemented in hardware. This week: what the bits mean, and how hardware adds.

Roadmap: Six Questions, One Thread

1. **Number systems** — how do we write numbers in binary, octal, hex — and convert?
2. **Signed representation** — how does hardware represent negative integers (two's complement)?
3. **Overflow and fixed point** — when does arithmetic fail, and how are fractions represented?
4. **Adder hardware** — how do we build a circuit that actually adds?
5. **Multiplication** — how does hardware multiply signed numbers without a separate sign step (Booth's algorithm)?
6. **Division** — how does hardware divide (restoring vs. non-restoring)?

Running thread for today

We will add $+7 + (-3)$ in binary — bit by bit — then build the circuit that does it, and check the result against the range of the format. The same adder, reused with control logic, is all multiplication and division need.

The bits are meaningless until we agree on what they mean.

Motivation: One Value, Many Alphabets

Socratic question

The decimal number 13 — how would you write it if you had only two symbols, or eight, or sixteen? Each is the *same* value, different alphabet.

- ▶ Decimal $13 = 1 \times 10^1 + 3 \times 10^0$.
- ▶ Binary: $1101_2 = 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 13$.
- ▶ Octal 15_8 , hex D_{16} — each a shorthand for groups of bits.
- ▶ Hardware stores *bits*; the base is just how we write them down.

Definition: Positional Number Systems

Positional notation

In base r , a number with digits $d_{n-1} \dots d_1 d_0$ has value

$$d_{n-1}r^{n-1} + \dots + d_1r^1 + d_0r^0.$$

The base is written as a subscript: 1101_2 , 15_8 , D_{16} .

- ▶ $r = 2$ binary (digits 0,1), $r = 8$ octal (0–7), $r = 10$ decimal, $r = 16$ hex (0–9, A–F).
- ▶ **Conversion**: repeated division by r gives the digits from least to most significant.
- ▶ Hex/octal are just binary in **groups**: 1 hex digit = 4 bits, 1 octal digit = 3 bits.

Worked Example: Decimal to Binary, and Back

$53_{10} \rightarrow$ binary (repeated division by 2)

$53/2 = 26 \text{ r } 1$; $26/2 = 13 \text{ r } 0$; $13/2 = 6 \text{ r } 1$; $6/2 = 3 \text{ r } 0$; $3/2 = 1 \text{ r } 1$; $1/2 = 0 \text{ r } 1$.
Read remainders bottom-up: 110101_2 .

Check: $110101_2 \rightarrow$ decimal

$1 \times 32 + 1 \times 16 + 0 \times 8 + 1 \times 4 + 0 \times 2 + 1 \times 1 = 32 + 16 + 4 + 1 = 53$. Correct.

Binary $\leftrightarrow$ hex is trivial by grouping: $11010101_2 = D5_{16}$ — no arithmetic needed, just 4-bit groups.

Socratic Check: How Many Bits?

Question

How many bits do you need to represent n distinct values? And with k bits, what is the largest unsigned integer?

- ▶ n values $\Rightarrow \lceil \log_2 n \rceil$ bits.
- ▶ k bits $\Rightarrow$ values 0 to $2^k - 1$ (unsigned).
- ▶ Example: 8 bits $\Rightarrow$ 0 to 255; 16 bits $\Rightarrow$ 0 to 65535.
- ▶ This ceiling will matter the moment we talk about **overflow**.

Unsigned is easy. The hard question: how do we represent negative numbers?

Three Candidate Schemes for Negatives

We have n bits. How do we encode a sign?

- ▶ **Sign-magnitude**: one sign bit + $n - 1$ magnitude bits. Simple, but two zeros ($+0$ and -0) and awkward addition.
- ▶ **One's complement**: negate = flip all bits. Still two zeros; addition needs end-around carry.
- ▶ **Two's complement**: negate = flip all bits and add 1. One zero, and *addition just works*.

Why two's complement wins

$-x$ is defined so that $x + (-x) = 0$ *in the same bit width*, with the carry out simply discarded. The ALU needs only an adder.

Definition: Two's Complement

Two's complement of an n -bit number

$$-x \equiv 2^n - x \pmod{2^n}.$$

Equivalently: flip all bits of x and add 1.

- ▶ Range: $-2^{n-1} \leq x \leq 2^{n-1} - 1$. For $n = 8$: -128 to $+127$.
- ▶ The **sign bit** is the MSB: 0 for non-negative, 1 for negative.
- ▶ Adding a number to its two's complement gives all zeros — the “self-cancelling” property that makes subtraction free.

(P&H, Sec. 3.2, p.188; Hamacher Ch. 2) [3, 1]

Worked Example: +7 and -3 in 8-bit Two's Complement

Represent both

+7 = 0000 0111₂. -3: flip 0000 0011 → 1111 1100, add 1 ⇒ 1111 1101₂.

Check by adding

0000 0111 + 1111 1101 = 1 0000 0100. Discard the carry-out (9th bit)
⇒ 0000 0100 = +4. $7 + (-3) = 4$.

The thread, started

This is the thread's first full example: negative numbers are ordinary bits, and **addition needs no special subtraction hardware** — just discard the carry-out.

Socratic Check: Negating Without a Calculator

Question

-5 in 8-bit two's complement is 1111 1011. Without computing anything, what is $-(-5)$, i.e. the two's complement of 1111 1011? What do you expect, and why?

- ▶ Flip 1111 1011 $\rightarrow$ 0000 0100, add 1 $\Rightarrow$ 0000 0101 = +5.
- ▶ Negating twice returns the original — two's complement is an **involution**:
 $-(-x) = x$.
- ▶ This self-inverse property is why negation is easy to implement and verify.

Two's complement makes negatives look like ordinary bits. But arithmetic can still overflow the range.

What Is Overflow?

Definition

Overflow occurs when the true result of an arithmetic operation cannot be represented in the format's range.

- ▶ Unsigned: carry out of the MSB is overflow.
- ▶ Two's complement: overflow $\neq$ carry out. A carry out of the MSB does *not* always mean overflow (recall $7 + (-3)$ above).
- ▶ The correct condition: overflow occurs when the sign of the result is *wrong* — i.e. adding two positives gives a negative, or two negatives give a positive.

(P&H, Sec. 3.2, p.188) [3]

Worked Example: Overflow vs. Carry

8-bit two's complement (range -128 to $+127$)

- ▶ $+100 + +50$: $0110\ 0100 + 0011\ 0010 = 1001\ 0110$, which reads as -106 — **wrong sign**, so **overflow**.
- ▶ $+127 + 1$: $0111\ 1111 + 1 = 1000\ 0000 = -128$ — again **wrong sign**, overflow.
- ▶ $-1 + (-1)$: $1111\ 1111 + 1111\ 1111 = 1\ 1111\ 1110$; discard carry
 $\Rightarrow 1111\ 1110 = -2$ — **correct**.

Rule of thumb

Overflow happens exactly when the two operands share a sign bit and the result's sign bit differs. Carry-out alone is *not* the test.

Fixed-Point Representation

Fractional numbers with a fixed binary point

A fixed-point number has an agreed position for the “binary point”:

$$b_{k-1} \dots b_0 . b_{-1} b_{-2} \dots b_{-m} \quad \text{value} = \sum_i b_i 2^i.$$

- ▶ E.g. Q8.8: 8 integer bits, 8 fraction bits — each fractional bit is $2^{-1}, 2^{-2}, \dots$
- ▶ $0.5 = 0.1_2$; $0.25 = 0.01_2$; $0.75 = 0.11_2$.
- ▶ Simple and fast (add/sub work on the same integer hardware), but the fixed point limits range and precision — which is why Week 3 moves to floating point.

(Hamacher Ch. 2; Mano Ch. 1) [1, 2]

Worked Example: A Fixed-Point Addition

Add $3.5 + 0.75$ in Q8.8

$3.5 = 0000\ 0011.1000\ 0000$; $0.75 = 0000\ 0000.1100\ 0000$.

Add bitwise

$0000\ 0011.1000\ 0000 + 0000\ 0000.1100\ 0000 = 0000\ 0100.0100\ 0000$. Read off: 4.25.

Correct: $3.5 + 0.75 = 4.25$.

The integer ALU adds fixed-point values unchanged — the programmer just agrees where the point is.

Socratic Check: When Does Fixed Point Fail?

Question

What happens if you try to store 0.1 in binary — does it terminate? What consequence does that have for fixed-point arithmetic?

- ▶ 0.1 in binary is a **repeating** fraction ($0.000110011\dots_2$) — it does not terminate.
- ▶ Fixed point must **truncate**, so 0.1 is stored approximately — repeated arithmetic accumulates error.
- ▶ This is the fundamental limitation that floating point (Week 3) manages but does not eliminate.

We can represent negatives and fractions. Now: the hardware that actually adds.

Motivation: From Addition to a Circuit

Socratic question

A human adds two numbers one column at a time, carrying. What is the smallest circuit that adds two *single bits* plus a carry-in — and how do we chain them into a multi-bit adder?

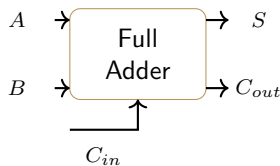
- ▶ **Half adder** (HA): two inputs A, B ; outputs sum S and carry C .
- ▶ **Full adder** (FA): three inputs A, B, C_{in} ; outputs sum S and carry C_{out} .
- ▶ Chain n full adders: each FA's C_{out} feeds the next C_{in} .

Definition: Half Adder and Full Adder

Boolean equations

Half adder: $S = A \oplus B$, $C = A \cdot B$.

Full adder: $S = A \oplus B \oplus C_{in}$, $C_{out} = A \cdot B + C_{in} \cdot (A \oplus B)$.

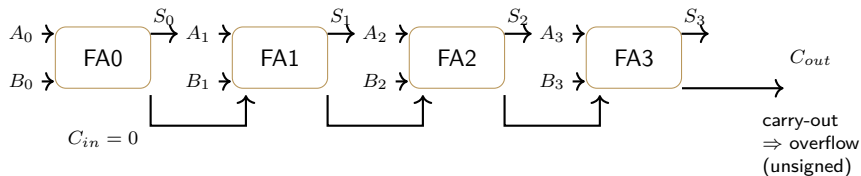


Worked Example: Full Adder Truth Table

A	B	C_{in}	S	C_{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Count the 1s in (A, B, C_{in}) : if odd, $S = 1$; if ≥ 2 , $C_{out} = 1$.

Ripple-Carry Adder: Chaining the Full Adders



The carry **ripples** from the least-significant to the most-significant bit — hence the name. The final C_{out} is the unsigned overflow flag.

Worked Example: Adding $7 + (-3)$ in Hardware

The thread, closed

$+7 = 0111$, $-3 = 1101$ (4-bit two's complement). Add with a 4-bit ripple-carry adder, $C_{in} = 0$:

Column by column

$0111 + 1101$: bit 0: $1 + 1 = 0$ carry 1; bit 1: $1 + 0 + 1 = 0$ carry 1; bit 2: $1 + 1 + 1 = 1$ carry 1; bit 3: $0 + 1 + 1 = 0$ carry 1. Result $0100 = +4$, carry-out = 1 (discarded). $7 + (-3) = 4$, no overflow — signs match the result.

One adder, same circuit, handles both signed and unsigned — the interpretation is the programmer's choice.

Socratic Check: Adder and Subtraction, One Circuit

Question

The ALU has only an adder. To compute $A - B$, what single extra step is needed — and what does it imply for the cost of subtraction hardware?

- ▶ Subtract B 's two's complement: $A - B = A + (-B)$.
- ▶ $-B = \text{flip } B + \text{add } 1$ — implementable with a MUX (choose B or $\bar{B}$) plus a forced carry-in of 1.
- ▶ So subtraction is **free**: one adder + a MUX, no separate subtractor needed. (P&H, Sec. 3.2, p.188) [3]

We have an adder. Now: how does hardware multiply — and divide?

Motivation: From Repeated Addition to a Faster Circuit

Socratic question

Multiplying $M \times Q$ by hand: add M to itself once per *set* bit of Q , shifting each time — the schoolbook shift-and-add. That always costs n additions for an n -bit Q , whether Q is $1111\dots$ or $0000\dots 1$. And it says nothing about *signed* operands.

- ▶ Shift-and-add treats the sign bit like any other bit — wrong for two's complement.
- ▶ We already have signed addition/subtraction (Act 4). Can we drive the *same* adder to multiply, without a special sign-handling step?
- ▶ **Booth's algorithm**: recode the multiplier so runs of 1s become one subtract and one add instead of many adds.

Definition: Booth's Recoding Rule

Registers (all n bits, M and Q hold the operands)

A (accumulator, init 0), Q (multiplier), Q_{-1} (one extra bit, init 0), M (multiplicand).
Repeat n times, examining (Q_0, Q_{-1}) each cycle:

- ▶ 00 or 11: no arithmetic — just arithmetic-shift-right (A, Q, Q_{-1}) by 1 bit.
- ▶ 01: $A \leftarrow A + M$, then arithmetic-shift-right.
- ▶ 10: $A \leftarrow A - M$, then arithmetic-shift-right.

After n cycles, the $2n$ -bit signed product is (A, Q) . (P&H, Sec. 3.3, p.191; Hamacher Ch. 6, Sec. 6.4.1, p.380) [3, 1]

Worked Example: $13 \times (-6)$ by Booth's Algorithm

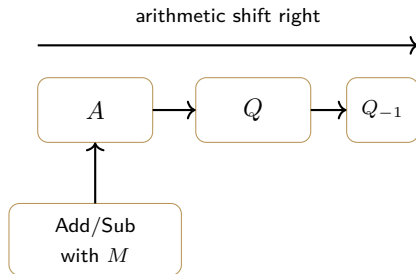
Setup (5-bit two's complement, $M = 13 = 01101$, $Q = -6 = 11010$)

Cycle	(Q_0, Q_{-1})	Operation	A after shift
1	(0, 0)	shift only	00000
2	(1, 0)	$A \leftarrow A - M$, shift	11001
3	(0, 1)	$A \leftarrow A + M$, shift	00011
4	(1, 0)	$A \leftarrow A - M$, shift	11011
5	(1, 1)	shift only	11101

Read off (A, Q)

$(A, Q) = 1110110010_2 = -78$. $13 \times (-6) = -78$. Correct.

Booth's Algorithm Hardware



$A \leftarrow A \pm M$ (or unchanged) based on (Q_0, Q_{-1}) , then the whole (A, Q, Q_{-1}) register shifts right by one bit — one adder, reused.

Socratic Check: Is Booth's Algorithm Ever Worse?

Question

Booth's algorithm always runs n cycles, same as plain shift-and-add. So where is the saving — and is there a multiplier pattern where Booth's does *no better* than shift-and-add?

- ▶ The saving is in **arithmetic operations**, not cycles: a run of consecutive 1s in Q becomes one subtract (at the run's start) and one add (just past its end) instead of one add per 1-bit.
- ▶ Worst case: alternating bits (0101...) — every cycle sees 01 or 10, so Booth's does an add/sub every cycle, same as shift-and-add.
- ▶ Booth's is never worse, and strictly better whenever Q has a run of 2 or more consecutive equal bits.

Multiplication reuses the adder. Division does too — but needs one more idea: the sign of the partial remainder.

Motivation: Division Is Subtraction, Repeated

Socratic question

Long division by hand: bring down a digit, subtract the divisor as many times as it fits, record the count, repeat. In binary, “as many times as it fits” collapses to *once or zero times per bit* — so division becomes shift, subtract, and a one-bit decision.

- ▶ Two hardware strategies differ only in *what happens when the subtraction goes negative*.
- ▶ **Restoring division**: subtract, and if the result is negative, add the divisor back (“restore”) before continuing.
- ▶ **Non-restoring division**: never restore — carry the negative partial remainder forward and flip the next operation instead.

Restoring Division

Registers (A : remainder, init 0; Q : dividend $\rightarrow$ quotient; M : divisor). Repeat n times:

1. Shift (A, Q) left by 1 bit.
2. $A \leftarrow A - M$.
3. If $A < 0$: **restore** $A \leftarrow A + M$ and set $Q_0 = 0$.
Else: set $Q_0 = 1$.

- After n cycles: Q = quotient, A = remainder.
- Every cycle does a subtract, and *half* the time (on average) an extra restoring add — up to 2 arithmetic operations per cycle.

(P&H, Sec. 3.4, p.197; Hamacher Ch. 6, Sec. 6.6, p.390-392) [3, 1]

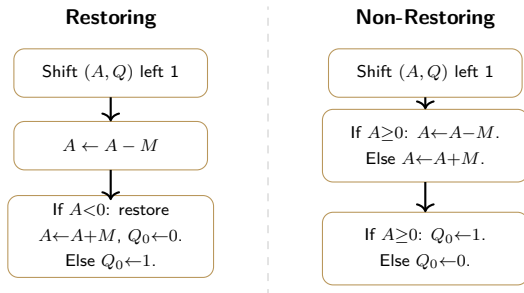
Non-Restoring Division

Same registers. Repeat n times:

1. Shift (A, Q) left by 1 bit.
2. If $A \geq 0$: $A \leftarrow A - M$. Else: $A \leftarrow A + M$.
3. If $A \geq 0$: set $Q_0 = 1$. Else: set $Q_0 = 0$.

- ▶ After n cycles: if $A < 0$, one final correction $A \leftarrow A + M$ recovers the true remainder. Q is already the correct quotient.
- ▶ Exactly *one* arithmetic operation per cycle, always — no restore step.

Restoring vs. Non-Restoring, Side by Side



Both repeat n times. Restoring can spend 2 arithmetic ops/cycle; non-restoring always spends exactly 1 (plus one final correction).

Worked Example: $13 \div 3$, Both Algorithms

4-bit unsigned, $Q = 13 = 1101$ (dividend), $M = 3 = 0011$ (divisor)

Both algorithms run 4 cycles and must agree on the answer: quotient $Q = 4 = 0100$, remainder $A = 1 = 0001$.

Check

$13 = 4 \times 3 + 1$. Quotient 4, remainder 1 — both algorithms converge on the same result; non-restoring just gets there with fewer arithmetic operations.

Full per-cycle traces are in this week's Lecture Notes and Instructor Handout.

Socratic Check: Why Non-Restoring Never Does More Work

Question

Restoring division does a subtract every cycle, plus a restoring add whenever $A < 0$. Non-restoring does exactly one add-or-subtract every cycle. Which does more total arithmetic operations, and why?

- ▶ Restoring: 1 op (subtract) + up to 1 more (restore) = up to 2 ops/cycle.
- ▶ Non-restoring: exactly 1 op/cycle, plus at most 1 final correction — never more than restoring, often fewer.
- ▶ Same quotient and remainder either way — non-restoring is a pure efficiency win, not a different algorithm.

We have the representation and the hardware — for addition, multiplication, and division. Now: put it together.

The Thread, Closed (I)

Representation and addition

- ▶ **Number systems**: positional notation; conversion by repeated division; hex/octal as bit grouping.
- ▶ **Two's complement**: $-x \equiv 2^n - x$; range -2^{n-1} to $2^{n-1} - 1$; negation = flip + 1.
- ▶ **Overflow**: wrong sign bit after addition of two same-signed operands — not the same as carry-out.
- ▶ **Fixed point**: agreed binary point; integer hardware works unchanged; fractions repeat in binary.
- ▶ **Adder**: HA $\rightarrow$ FA $\rightarrow$ ripple-carry; $A - B = A + (-B)$ is free.

The running thread

$+7 + (-3)$: represent both in two's complement, add with a ripple-carry adder, discard the carry-out, check the sign — 4, no overflow.

The Thread, Closed (II)

Multiplication and division: the same adder, reused

- ▶ **Booth's multiplication**: examine (Q_0, Q_{-1}) — add/ subtract/none, then arithmetic-shift-right — handles signed operands with the same adder, no special sign step.
- ▶ **Division**: restoring (subtract, test, restore-if-negative) vs. non-restoring (conditional add/subtract, no restore) — same quotient and remainder, non-restoring never does more work.

The thread, extended

The same adder, driven by different control logic, multiplies $13 \times (-6) = -78$ (Booth's) and divides $13 \div 3 = 4$ remainder 1 (restoring/non-restoring) — no new arithmetic hardware needed.

Bridge to Week 3

What's still missing

We now have all four integer operations (add, subtract, multiply, divide) in hardware — but fixed point handles fractions only with fixed range/precision. Real numbers span enormous ranges (from 10^{-300} to 10^{300} in science) — one fixed point cannot cover them.

Next week

Floating point: the IEEE 754 representation — sign, exponent, mantissa — trading a little precision for a huge range. Week 3 builds the format, the normalization, and the arithmetic.

Summary

- ▶ **Number systems:** $\text{value} = \sum d_i r^i$; convert by repeated division; hex/octal group bits.
- ▶ **Two's complement:** $-2^{n-1} \leq x \leq 2^{n-1} - 1$; negate by flip + 1; addition needs no special subtraction hardware.
- ▶ **Overflow:** carry-out $\neq$ overflow; wrong result sign is the test.
- ▶ **Fixed point:** an agreed binary point; simple but limited range and precision.
- ▶ **Adder:** HA/FA equations; ripple-carry chain; $A - B = A + (-B)$.
- ▶ **Booth's algorithm:** (Q_0, Q_{-1}) recoding — 01 adds, 10 subtracts, 00/11 shift only; never worse than shift-and-add.
- ▶ **Division:** restoring subtracts then conditionally restores (up to 2 ops/cycle); non-restoring always does exactly 1 op/cycle.

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